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# Paleomagnetism of igneous rocks from the Belcher Islands, Northwest Territories, Canada[1]

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Paleomagnetic results from igneous rock units on the Belcher Islands, Hudson Bay, are described. Fold tests for all units studied, as well as a contact test for the intrusive bodies, indicate that both primary (initial), and secondary (post-folding) magnetizations are present.

The paleomagnetic pole position from primary directions of the oldest unit studied, the Eskimo volcanics, is situated at 40°S, 002°E ($A_{95}$ = 12°) and is similar to that derived from equivalent volcanics on the mainland. The younger volcanic unit studied, the Flaherty volcanics, yielded a pole position from primary directions at 0°, 244°E ($A_{95}$ = 7°). The Haig intrusions, associated with these younger volcanics, yields an almost identical pole position at 1°N, 247°E ($A_{95}$ = 6°), being derived from directions which are shown to be not only pre-folding but also date from initial cooling. The Eskimo volcanics, which have been more deeply buried than the Flaherty (upper) volcanics, carry substantial components of secondary (post-folding) magnetization which yield a pole position at 19°N, 243°E ($A_{95}$ = 15°), about 20° north of the pole positions derived from the youngest units.

It is argued that the apparent polar wander path (APWP) constructed for the Belcher Islands is representative of the mainland Ungava Craton. Comparison with the equivalent APWP from elsewhere in North America shows that the two APWP's are at variance. Although a two-plate model could be advanced, perhaps a more conservative interpretation is to extend the existing North American APWP eastward to include the Belcher–Ungava APWP, that is, to favour a one-plate model.

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On décrit les résultats d'études paléomagnétiques sur les unités de roches ignées des îles Belcher, dans la baie d'Hudson. Des vérifications pour le plissement dans toutes les unités étudiées de même qu'un essai de contact pour les massifs intrusifs indiquent la présence d'aimantations primaires (initiales) et secondaires (après le plissement).

La position du pôle paléomagnétique pour les directions primaires de l'unité la plus ancienne qu'on a étudiée, les volcaniques d'Eskimo, se situe à 40°S, 002°E ($A_{95}$ = 12°) et est semblable à celle qu'on mesure dans les roches volcaniques équivalentes sur le continent. Pour l'unité volcanique la plus récente qu'on a étudiée, les volcaniques de Flaherty, la direction du pôle à partir des directions primaires se situe à 0°, 244°E ($A_{95}$ = 7°). Les intrusions de Haig, associées à ces volcaniques récentes, indiquent une position de pôle à peu près identique à 1°N, 247°E ($A_{95}$ = 6°) à partir de directions d'aimantation qu'on peut faire remonter avant le plissement jusqu'au moment du refroidissement initial. Les volcaniques d'Eskimo, qui ont été enfouies plus profondément que les volcaniques (supérieures) de Flaherty, renferment des composantes d'aimantation secondaire substantielles (après le plissement) qui donnent une position de pôle à 19°N, 243°E ($A_{95}$ = 15°), environ 20° au nord des positions de pôles provenant des unités plus récentes.

On propose que le trajet apparent du cheminement polaire établi pour les îles Belcher représente le craton d'Ungava sur le continent. La comparaison avec le trajet équivalent pour

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d'autres endroits en Amérique du Nord montre que les deux trajets sont en opposition. Bien qu'on puisse proposer un modèle à deux plaques, il est probable qu'une interprétation plus raisonnable serait d'étendre le trajet nord-américain apparent vers l'est pour inclure le trajet apparent de Belcher–Ungava, c'est-à-dire de favoriser un modèle à une plaque.

[Traduit par le journal]

## Introduction

The Belcher Islands in southeastern Hudson Bay, Canada, form part of the circum-Ungava geosyncline. During Proterozoic times this geosyncline is thought to have surrounded (and overlaid the margins of) the Superior Province of Quebec (the Ungava Craton). Remnants of the geosyncline are preserved as the Labrador Trough, the Cape Smith Fold Belt, and the Belcher Fold Belt, all of which consist of cyclic volcanogenic and sedimentary horizons. The stratigraphy and structure of the Belcher Islands are well established and they, therefore, represent excellent paleomagnetic targets since little is known, with certainty, of the origin or paleomagnetism of this geosyncline.

Previous paleomagnetic investigations of the geosyncline include that by Fujiwara and Schwarz (1975) of the Cape Smith basalts, that of the Richmond Gulf area by Schwarz (1976), and a preliminary investigation by Park (1977) of the Labrador Trough. These are discussed in relation to the findings herein towards the end of the paper.

## Geology

Jackson (1960) has recognized three igneous units on the Belcher Islands, two of which are extrusive and another which is intrusive. The oldest igneous unit, the Eskimo Formation, is comprised of up to about 1000 m of basalts (and some interbedded sediments) outcropping as inliers in the cores of the main islands (Fig. 1). The next igneous unit, the Flaherty Formation, is separated from the Eskimo Formation by a thick sedimentary pile and consists of up to 2000 m of basalts and intervening sedimentary–pyroclastic sequences. The Haig intrusions, which are medium-grained diabases, intrude all units below, but none above, the Flaherty Formation, and are "probably related to the volcanic activity represented by the Flaherty Formation" (Dimroth *et al.* 1970). Another minor intrusive unit is apparently much younger than the other units of the Belcher Islands and has not been included in the present study (Hoffmann and Jackson 1969).

During the completion of infilling of the basin, a major deformation and mild metamorphism occurred, resulting in the structures and relationships presently exposed on the island. This deformation took place 1.6–1.7 Ga ago and has been correlated

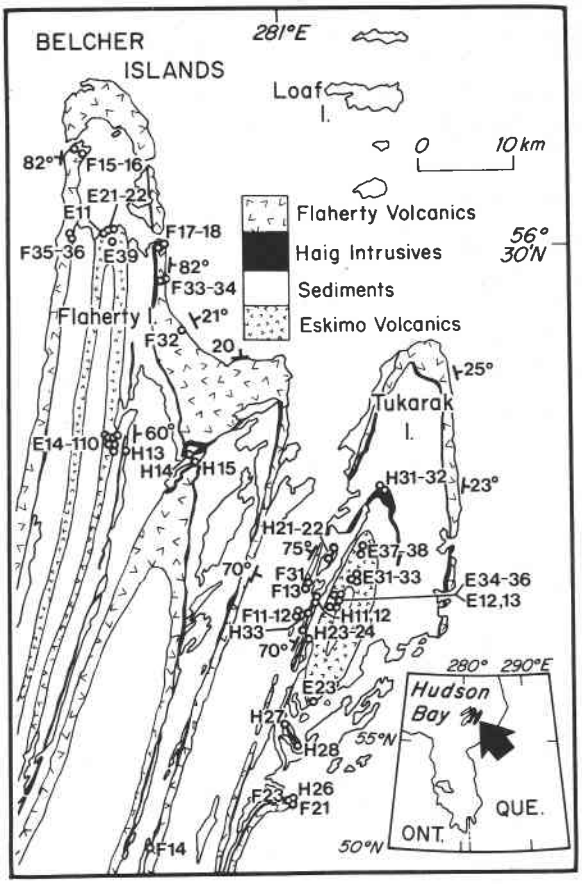



FIG. 1. Geology and site localities. The letters E, F, and H refer to the Eskimo and Flaherty volcanics and the Haig intrusives respectively. Numerals 1, 2, or 3 immediately following a letter indicate that the site was collected during the first, second, or third field season respectively. The remaining numeral(s) are the site numbers.

with the Hudsonian orogeny. The metamorphic facies of the volcanics has been determined as subgreenschist (Stirby 1975).

Jackson (1967) reports six whole-rock K–Ar ages from igneous rocks of the Belcher Islands, 1.625 ± 0.210 Ga from the Eskimo volcanics, 1.693 ± 0.230 Ga from the Flaherty volcanics, 1.620 Ga (no error stated) from the Haig intrusives, and three somewhat younger ages (ca. 900 Ma). He interprets the older ages as approximate ages of the main igneous activity and the younger ages as being associated with the young intrusives mentioned above. Fryer (1972) gives two Rb–Sr isochrons: one for the Flaherty (upper) volcanics and as-

sociated Haig intrusives and another for some overlying slates. These yield ages of 1.790 ± 0.089 Ga and 1.755 ± 0.069 Ga respectively ($^{87}Rb$ decay constant of $1.39 \times 10^{-11}$ year$^{-1}$). Initial $^{87}Sr/^{86}Sr$ ratios are 0.706. Brooks (unpublished report to the Earth Physics Branch) represents Rb/Sr isochrons for the Eskimo (lower) volcanics giving an age of 1.750 ± 0.025 Ga with an initial $^{87}Sr/^{86}Sr$ of 0.705 and for the Flaherty (upper) volcanics giving an age of 1.640 ± 0.065 Ga ($^{87}Rb$ decay constant of $1.39 \times 10^{-11}$ year$^{-1}$). The sequence, therefore, appears to bracket 1.7 Ga and for the purposes of this report ages of 1.75 and 1.64 Ga are assigned to the Eskimo and Flaherty volcanics respectively.

## Sampling and Techniques

Three separate field trips have been undertaken and a total of 301 oriented samples from 52 sites in igneous units (plus baked contacts) have been collected (Fig. 1).

Throughout each collection portable rock drills have been used to obtain samples (cores) while magnetic and solar (whenever possible) methods have been used to orient the samples. Measurements were performed on Schonstedt magnetometers (SSM-1 and DSM-1) and demagnetization was effected using the furnace unit described by Roy *et al.* (1971), Schonstedt TSM furnaces, and the alternating field (AF) demagnetizer described by Roy *et al.* (1972).

Firstly, all specimens have been treated as "pilots" since their salient characteristics proved to be independent of site, and in most cases, of sample. Several specimens from each site have been subjected routinely to stepwise thermal and alternating field demagnetization procedures. Between 7 and 18 steps were used for each specimen depending upon their intensities; the limit of measurement being about 0.1 mA m$^{-1}$ ($10^{-7}$ emu/cm$^3$). Overall, thermal demagnetization proved to be more successful than AF demagnetization.

Secondly, many measurements were rejected because, although their intensities were well above the level of sensitivity, a scrutiny of the measured components often revealed that various portions of specimens were magnetized along different directions. During a measurement of the magnetization of a specimen, four separate readings of each orthogonal component are taken, i.e., the magnetization in each plane is measured twice. Built into the Schonstedt DSM-1 program are angular parameters which reflect the homogeneity of magnetization. These parameters are the differences between the two directions of magnetization measured for each plane. It was found that using a 20° cutoff was sufficient to sort measurements into a homogenous class, in which case the measurements were further processed, or into an inhomogenous class. Some specimens were sliced, using a diamond saw, in an attempt to isolate these uni-magnetized pockets. This, however, proved to be unsuccessful. The reasons for these difficulties are not entirely clear but, as discussed below, opposed (almost reversed) magnetizations appear to be common and must rate high as a cause.

Lastly, the use of vector diagrams has been an invaluable aid in investigating the demagnetization behaviour of these rocks. In particular, opposed magnetic components are easily recognised and separated.

Isolated magnetic components are recognised as linear segments consisting of three or more consecutive colinear end points. This basis has been used throughout this study as a simple objective criterion. Only results from specimens which reveal at least one such set of colinear points have been used to calculate mean directions of samples, and only sites for which results from at least two samples are consistent have been used to calculate mean formation directions.

## Results

No broad generalizations can be stated which adequately describe the demagnetization behaviour of these rocks. As mentioned above, even specimens from the same sample have responded differently to partial demagnetization, and, indeed, individual portions within the same specimen appear to have behaved differently. The salient features of each formation, beginning with the oldest, are given below. As a convention, directions with positive inclinations are referred to as normal while those with negative inclinations are referred to as reversed, and primary directions are defined as being those that are pre-folding while secondary directions are post-folding.

### *Eskimo Volcanics*

The natural remanent magnetization (NRM) intensities varied from less than 1 mA m$^{-1}$ to greater than 14 A m$^{-1}$; however, no correlation of intensity and demagnetization characteristics appeared to be significant. The NRM directions were scattered on the lower hemisphere. The most striking feature of these specimens was their inhomogeneity of magnetization. Representative demagnetization diagrams are given in Fig. 2. It is instructive to note that the components initially removed from specimens 39A and 40A (site E35) were opposed to the component which was isolated and eventually

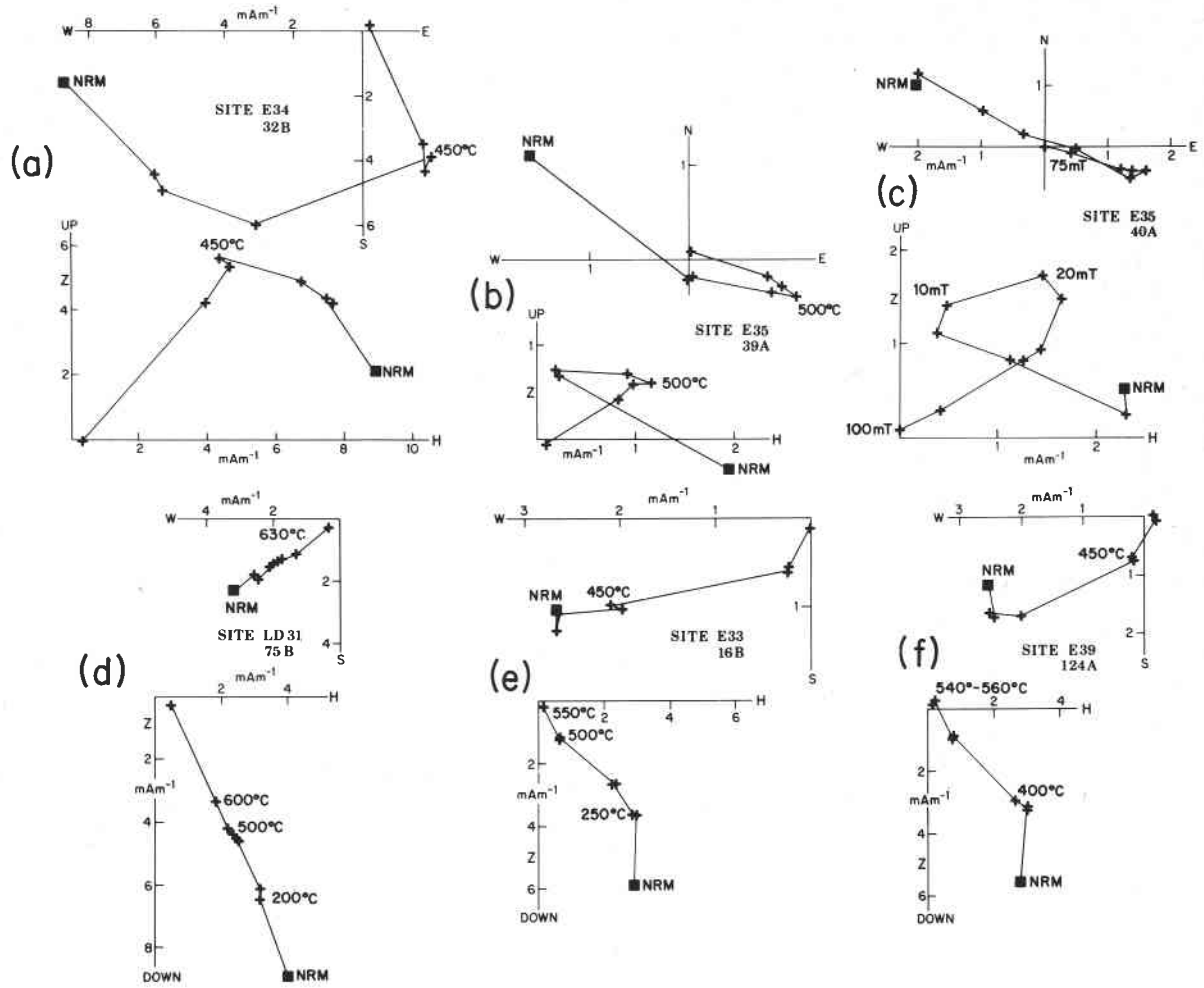



FIG. 2. Vector diagrams displaying demagnetization behaviour of samples from the Eskimo volcanics. For each data point the vertical component has been plotted against the total horizontal component (Roy and Park 1974). Declination, inclination, and intensity can be read directly from the plots.

completely demagnetized after further treatment (using both AF and thermal methods). On the basis of these and other results described below, it is suggested that opposed (almost antiparallel) components might be the reason for the inhomogeneously magnetized specimens encountered. No simple normal magnetizations were found in the Eskimo volcanics, as they were in the younger formations, so it is possible that these inhomogeneous magnetizations are not antiparallel, perhaps being composed of a slightly oblique secondary and a primary direction giving the appearance of antiparallel components. Some samples do indeed yield a simple secondary component (Fig. 2*e* and below).

The median destructive fields (MDF) observed were invariably below 50 mT, and unblocking temperatures always below 600°C which is taken to indicate that titanomagnetite is the carrier of the remanences in the Eskimo volcanics. The primary directions identified are thought to be original and associated with initial thermoremanent magnetization (TRM), but the existence of antiparallel magnetic components suggests the possibility of chemical remanent magnetizations (CRM), maybe related to deuteric and (or) later low temperature alteration. This is discussed in more detail later. Seven sites out of 15 have yielded consistent directions after correcting for folding (Fig. 7*a*).

The increase in $k$ (McElhinny 1964) after application of the Graham (1949) fold test is significant at 99% and shows that these directions are pre-folding beyond reasonable doubt.

The mean site direction for the primary magneti-

TABLE 1. Eskimo volcanics

| Site (Locality) | Bedding (DDA° | Dip°) | TR | Directions $D_H$° | $I_H$° | $D_B$° | $I_B$° | $N$ | $R$ | $k$ | $\alpha_{95}$° | Poles Lat.(°N) | Long.(°E) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| E12 | 292 | 70 | 500–560°C | 269 | −61 | 128 | −46 | 2 | 1.97 | — | — | −43 | 355 |
| E18 | 095 | 80 | 580°C, 20 mT | 105 | 46 | 103 | −33 | 4 | 3.98 | 147 | 8 | −22 | 010 |
| E21 | 354 | 10 | 450–550°C 7.5–40 mT | 100 | −53 | 111 | −49 | 5 | 4.95 | 87 | 8 | −36 | 012 |
| E32 | 080 | 12 | 500–525°C 60–80 mT | 112 | −39 | 119 | −49 | 4 | 3.99 | 464 | 4 | −41 | 006 |
| E34 | 295 | 49 | 500–525°C 25–30 mT | 253 | −60 | 154 | −58 | 3 | 2.96 | 51 | 18 | −66 | 336 |
| E35 | 295 | 52 | 375–525°C 10–25 mT | 093 | −88 | 114 | −36 | 3 | 2.95 | 43 | 19 | −30 | 002 |
| E39 | 025 | 25 | 530–575°C | 097 | −40 | 120 | −43 | 3 | 2.98 | 123 | 11 | −37 | 000 |
| Mean | | | | 114 | −64 | 119 | −46 | | | | | −40 | 002 |
| $k$ | | | | 2.5 | | 36 | | | | | | 27 | |
| $\alpha_{95}$° | | | | 49 | | 10 | | | | | | ($A_{95}$°) 12 | |

NOTES: Symbols: DDA° = downdip azimuth; TR = treatment range yielding isolated directions; $D_H$°, $I_H$°, $D_B$°, $I_B$° = declination and inclination with respect to present horizontal and bedding respectively; $N$, $R$, $k$, $\alpha_{95}$° = number of samples or sites, length of resultant of $N$ unit vectors, precision parameter, and half-angle of the cone of confidence respectively (Fisher 1953). Poles are based on directions with respect to bedding (see text).

zation of the Eskimo volcanics is 119°, −46°, $N = 7$, and $\alpha_{95} = 10°$ and the mean site virtual geomagnetic pole (VGP) is at 40°S, 002°E with $A_{95} = 12°$ (Table 1).

*Overprint*

A secondary magnetization from seven sites has been observed. This magnetization was sometimes the only component present (Fig. 2*d*,*e*) but was occasionally found to coexist with a primary direction. Specimen 124A from site E39 yielded directions of about 235°, +50° (referred to the present horizontal) between 200 and 400°C, and 130°, −50° (after unfolding) between 530 and 575°C (Fig. 3). These directions are not significantly different from the mean secondary (overprint) direction given in Table 2 and the mean primary direction given in Table 1 respectively. The fold test, based on site mean directions, is negative, showing that the directions are post-folding.

The mean site overprint direction is 234°, +60°, $N = 7$, and $\alpha_{95} = 12°$ and the mean VGP position is 19°N, 243°E with $A_{95} = 15°$. This pole position is not significantly different from the pole position reported by Fujiwara and Schwarz (1975) for the Cape Smith basalts, uncorrected for bedding tilt. In that study it was not possible to determine whether the magnetizations were pre- or post-folding and consequently no definitive interpretation could be made. In the light of the secondary directions found here, and their agreement with those from Cape Smith, the pole position from the Cape Smith basalts is correlated with a post-folding (post-Hudsonian) age. This was the preferred interpretation given by Fujiwara and Schwarz (1975).

Out of 97 samples 49 have not yielded well defined primary (pre-folding) magnetizations or secondary (post-folding) magnetizations. The possibility of opposed (antiparallel) components which could not be separated using the existing cleaning techniques may be the reason for this.

*Flaherty Volcanics*

The NRM intensities varied between less than 1 mA $m^{-1}$ to over 2.5 A $m^{-1}$. These were generally scattered on the lower hemisphere, although some had steep negative inclinations. There did not appear to be any correlation between NRM intensity and demagnetization characteristics, the most prevalent feature being magnetic inhomogeneity. Magnetically opposed components were commonly encountered and in many cases they have been isolated from the same specimen. Specimen 15B from site F12 showed a reversal in polarization between 450 and 530°C while specimen 152B from site F15 revealed a series of such reversals between 500 and 570°C (Fig. 4*b*). Other specimens (120B and 122B from site F36), after cleaning, gave an apparently single magnetization of normal polarity (approximately 220°, + 50°). No unblocking temperatures were observed above 600°C and MDFs were generally less than 20 mT, indicating titanomagnetite is probably the main remanence carrier.

Samples from 11 out of 14 sites have yielded directions, after thermal or AF treatment and bedding correction, either northeast and up or south-

TABLE 2. Eskimo overprint

| Site (Locality) | Bedding (DDA° | Dip°) | TR | Directions $D_H$° | $I_H$° | $D_B$° | $I_B$° | $N$ | $R$ | $k$ | $\alpha_{95}$° | Poles Lat.(°N) | Long.(°E) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| E12 | 292 | 70 | 400–500°C | 217 | 53 | 256 | 07 | 3 | 2.99 | 187 | 9 | 05 | 251 |
| E14 | 095 | 80 | 400–450°C | 249 | 60 | 111 | 36 | 3 | 2.88 | 17 | 31 | 23 | 231 |
| E15 | 095 | 80 | 450°C | 262 | 39 | 115 | 59 | 2 | 1.99 | — | — | 14 | 210 |
| E16 | 095 | 80 | 530–560°C | 220 | 74 | 109 | 19 | 3 | 2.96 | 54 | 17 | 32 | 259 |
| E23 | 170 | 12 | 500°C | 214 | 75 | 195 | 65 | 3 | 2.99 | 354 | 7 | 31 | 262 |
| E33 | 080 | 12 | 375°C, 7.5 mT | 225 | 58 | 209 | 67 | 4 | 3.99 | 322 | 5 | 12 | 246 |
| E39 | 025 | 25 | 200°C | 227 | 55 | 263 | 75 | 5 | 4.96 | 88 | 8 | 10 | 243 |
| Mean | | | | 234 | 60 | 162 | 69 | 7 | | | | 19 | 243 |
| $k$ | | | | 27 | | 3.1 | | | | | | 17 | |
| $\alpha_{95}$° | | | | 12 | | 41 | | | | | | ($A_{95}$°) 17 | |

NOTES: Symbols: See Table 1. The poles are calculated from the directions with respect to present horizontal.

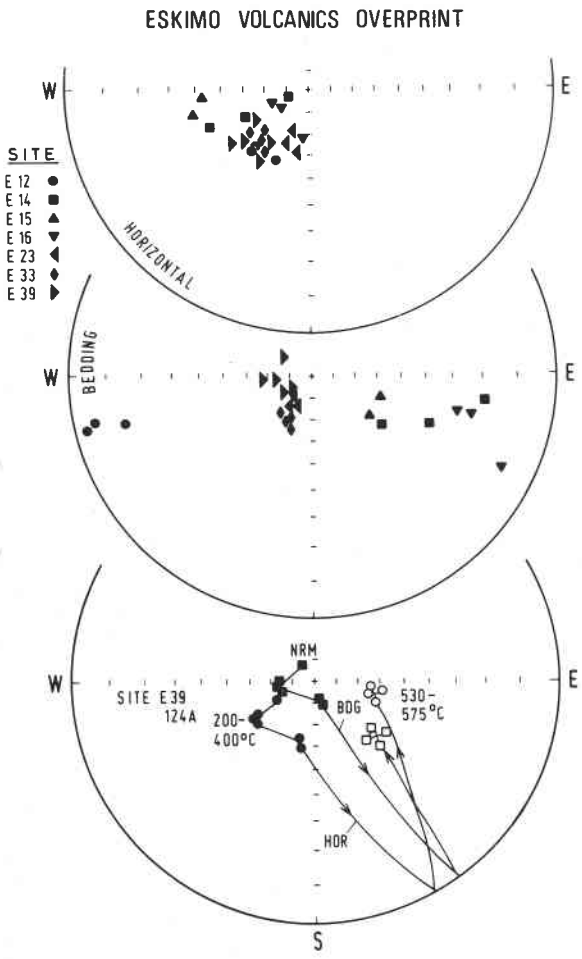



FIG. 3. Stereographic projections comparing secondary directions of magnetizations (overprints) with respect to the present horizontal and the bedding planes for samples of the Eskimo volcanics. Site mean directions are given in Table 2. The bottom stereograph displays the change in the direction of magnetization from 25–575°C for specimen 124A from site E39, with respect to present horizontal and with respect to the bedding plane.

west and down, and in many cases both directions were observed in single samples. The increase in $k$ after correction for bedding is a factor of 15 which is significant at 99% (McElhinny 1964), and shows that these directions are related to pre-folding magnetizations (Fig. 7*b*). The mean site direction, taking into account the opposite polarities, is 042°, −44°, $N = 11$, and $\alpha_{95} = 7°$, and the mean site VGP is at 0°N, 244°E with $A_{95} = 7°$ (Table 3). There did not appear to be any systematic magnetic directions with respect to the present horizontal. Only after corrections for bedding was a characteristic direction recognized which is interpreted to indicate that these rocks have not suffered significant post-folding (secondary) remagnetization, and have apparently escaped the event(s) that affected the Eskimo volcanics.

Of the 93 samples collected from the Flaherty (upper) volcanics 58 have not yielded useful results. Rather than inherent magnetic instability, the intimate association of opposed components within the same specimen (Fig. 4) is thought to be the main reason for this yield because the vast majority of these specimens were very inhomogeneously magnetized.

*Haig Intrusions*

The NRM directions showed two groupings. One was steeply positive inclined in the southwest quadrant while the other was shallow and negatively inclined in the northeast quadrant. This is evident before and after correction for bedding tilt because the majority of sites were collected from sills with shallow dips. As is shown after treatment (Fig. 7*c*), the reason for these two NRM groupings was the effect of reversed primary magnetizations overprinted by secondary, perhaps viscous remanent magnetizations (VRM) in the present or recent field direction, constituting a reversal test (Irving 1964; McElhinny 1973). It is satisfying to note that

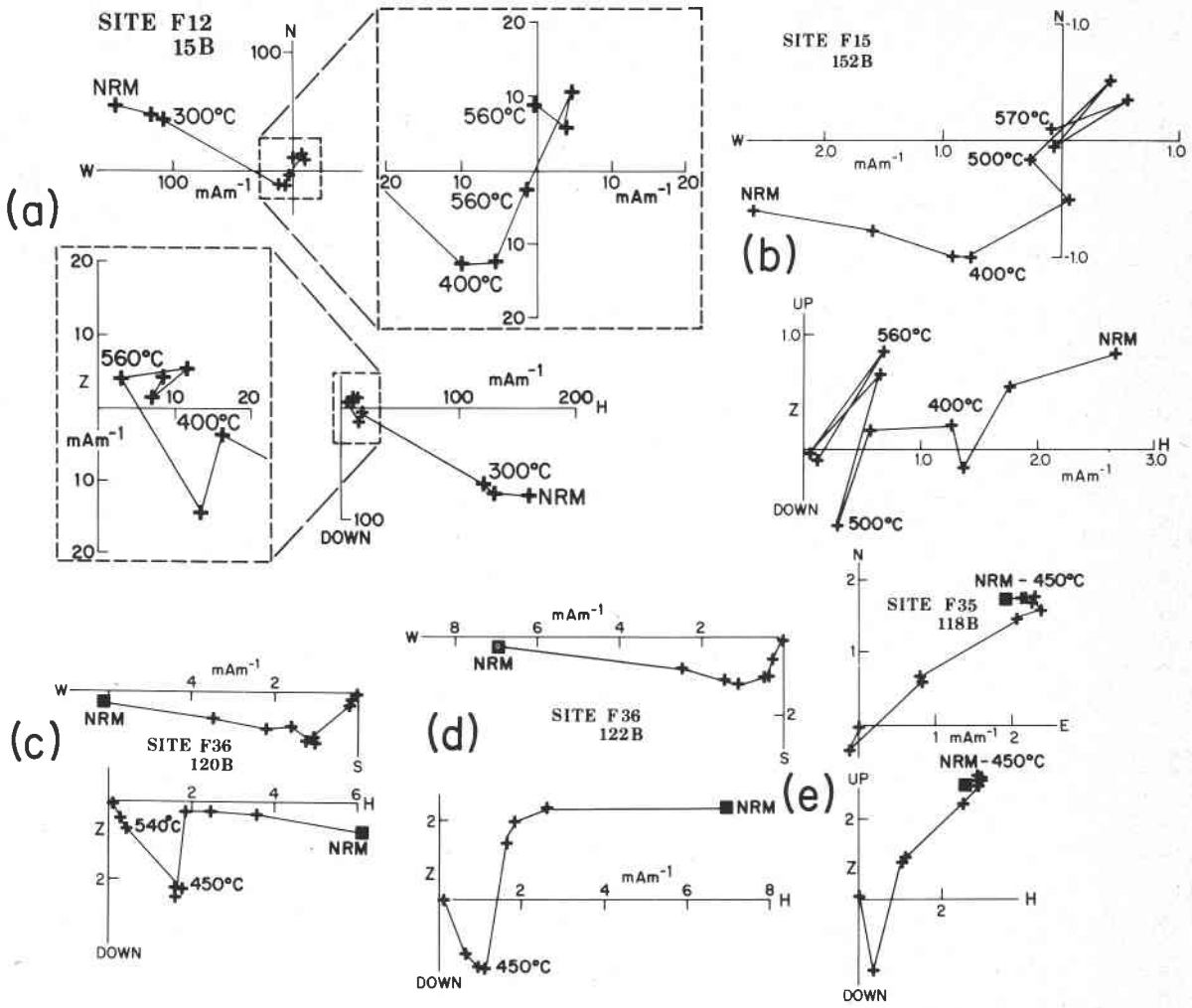



FIG. 4. Vector diagrams displaying demagnetization behaviour of samples from the Flaherty volcanics. See Fig. 2 for explanation.

the result by applying the reversal test is the same as that after extensive laboratory treatment (although the error is much smaller in the latter).

The NRM intensities showed a much smaller range (≈2–280 mA m$^{-1}$) than those of the volcanics. Except for samples from a fine-grained chilled contact site (H31) most NRM intensities were below 50 mA m$^{-1}$. The magnetizations of these rocks were more homogeneous than those of the Eskimo and Flaherty volcanics, and generally showed simpler demagnetization characteristics, although all specimens from one site (H13) increased in intensity between 540 and 550°C. This was sometimes accompanied by a small change of direction (10–30°), but on other occasions no direction change occurred (specimen 55A, Fig. 5*a*), indicating that the removed component was, in these cases, *exactly* reversed.

Representative demagnetization characteristics are presented in Fig. 5. Maximum MDFs of about 100 mT and unblocking temperatures of about 570°C are present in, but not typical of, the diabase. Specimen 20B (site H31) revealed sharp unblocking temperatures between 450 and 500°C, probably indicating the dominance of impure magnetite as the remanence carrier. As with the volcanics the absence of coercivities of remanence in excess of 100 mT and unblocking temperatures in excess of 600°C as a general feature of the diabase, is taken to indicate that titanomagnetite is the dominant remanence carrier throughout the diabase.

Table 4 gives the site mean directions before and after correction for bedding. It is evident that the directions agree more closely when referred to their bedding planes (Fig. 7*c*), with the precision ($k$) increasing by a factor of 24, representing a positive

TABLE 3. Flaherty volcanics

| Site (Locality) | Bedding (DDA° | Dip°) | TR | Directions $D_H$° | $I_H$° | $D_B$° | $I_B$° | $N$ | $R$ | $k$ | $\alpha_{95}$° | Poles Lat.(°N) | Long.(°E) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| F13 | 285 | 78 | 550–560°C | 159 | 43 | 239 | 34 | 3 | 2.96 | 49 | 18 | −01 | 227 |
| F14 | 289 | 55 | 300–400°C 15–20 mT | 000 | −35 | 041 | −33 | 2 | 1.96 | — | — | −08 | 242 |
| F15 | 295 | 85 | 400–560°C 40–100 mT | 339 | −15 | 041 | −46 | 4 | 3.92 | 40 | 15 | 01 | 245 |
| F16 | 295 | 85 | 400–570°C | 336 | −14 | 041 | −49 | 3 | 2.92 | 25 | 25 | 04 | 246 |
| F17 | 086 | 85 | NRM–570°C | 058 | 31 | 051 | −45 | 5 | 4.85 | 26 | 15 | 04 | 237 |
| F18 | 086 | 85 | 300–540°C 5–20 mT | 055 | 11 | 024 | −55 | 3 | 2.98 | 93 | 13 | 04 | 261 |
| F21 | 250 | 18 | 500°C 20–25 mT | 030 | −47 | 039 | −32 | 2 | 1.99 | — | — | −10 | 244 |
| F31 | 290 | 60 | 450–565°C | 355 | −22 | 028 | −32 | 3 | 2.92 | 25 | 26 | −13 | 254 |
| F34 | 080 | 64 | NRM–565°C 40 mT | 062 | 07 | 050 | −53 | 3 | 2.96 | 47 | 18 | 09 | 240 |
| F35 | 270 | 80 | 300–540°C 5–20 mT | 307 | −34 | 041 | −49 | 4 | 3.98 | 185 | 7 | 03 | 246 |
| F36 | 270 | 80 | 500–525°C | 117 | −32 | 225 | 57 | 3 | 2.96 | 46 | 18 | 11 | 246 |
| Mean | | | | 002 | −24 | 042 | −44 | 11 | | | | 00 | 244 |
| $k$ | | | | 3.2 | | 46 | | | | | | 49 | |
| $\alpha_{95}$° | | | | 31 | | 7 | | | | | | ($A_{95}$°) 7 | |

NOTES: Symbols: See Table 1. The poles are calculated from directions with respect to bedding.

TABLE 4. Haig intrusives

| Site (Locality) | Bedding (DDA° | Dip°) | TR | Directions $D_H$° | $I_H$° | $D_B$° | $I_B$° | $N$ | $R$ | $k$ | $\alpha_{95}$° | Poles Lat.(°N) | Long.(°E) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| H11 | 270 | 70 | 450–550°C | 307 | −33 | 029 | −55 | 4 | 3.86 | 21 | 21 | 05 | 258 |
| H12* | 270 | 70 | 300–525°C | 314 | −27 | 018 | −49 | 2 | 1.99 | — | — | −03 | 266 |
| H13 | 099 | 75 | 400–560°C | 060 | 10 | 040 | −44 | 6 | 5.96 | 140 | 6 | −01 | 245 |
| H15 | Horizontal | | 300–575°C | 221 | 45 | 221 | 45 | 4 | 3.95 | 57 | 12 | −01 | 245 |
| H21 | 291 | 62 | 500–540°C | 160 | 33 | 221 | 48 | 2 | 1.95 | — | — | 02 | 247 |
| H26 | 201 | 17 | 570–580°C | 046 | −46 | 039 | −30 | 3 | 2.99 | 132 | 11 | −11 | 243 |
| H27 | 255 | 16 | 500–560°C | 214 | 68 | 230 | 54 | 5 | 4.97 | 143 | 6 | 10 | 241 |
| H28 | 245 | 21 | 500–540°C | 206 | 65 | 222 | 47 | 4 | 3.95 | 66 | 11 | 01 | 245 |
| H31 | 058 | 14 | 375–500°C 10–25 mT | 046 | −27 | 044 | −41 | 6 | 5.99 | 375 | 4 | −02 | 241 |
| H32 | 058 | 14 | 450–525°C 10–30 mT | 052 | −34 | 050 | −48 | 5 | 4.95 | 85 | 8 | 05 | 239 |
| Mean | | | | 20 | −45 | 040 | −46 | 10 | | | | 01 | 247 |
| $k$ | | | | 4.4 | | 23 | | | | | | 66 | |
| $\alpha_{95}$° | | | | 26 | | 6 | | | | | | ($A_{95}$°) 6 | |

NOTES: Symbols: See Table 1. The poles are calculated from directions with respect to bedding.
*This site is in the baked contact zone near site H11.

fold test with 99% confidence (McElhinny 1964). The overall mean site direction, with respect to the bedding planes, is 040°, −46°, $N = 10$, and $\alpha_{95} = 6°$, while the mean VGP position is 1°N, 248°E ($A_{95} = 6°$).

Samples from a baked contact (site H12) yielded magnetizations which are indistinguishable from those of the diabase (site H11) — Fig. 6 shows the change in direction and intensity of specimens 34B (site H11) and 38A (site H12). Demagnetization

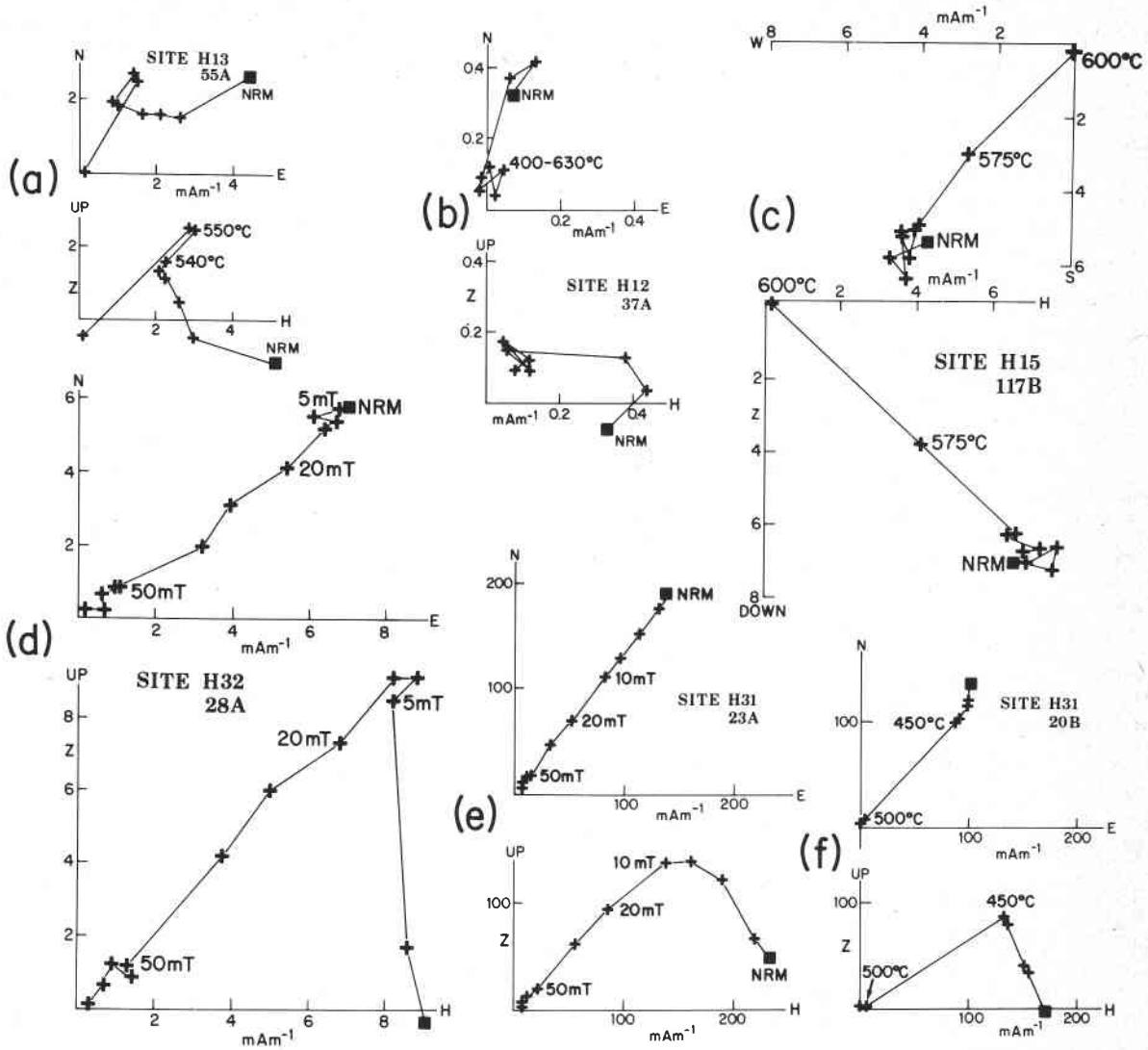



FIG. 5. Vector diagrams displaying demagnetization behaviour of samples from the Haig intrusives. See Fig. 2 for explanation.

characteristics of another baked contact specimen are given in Fig. 5*b* (specimen 37A, site H12). The intensities of these baked sediments were weak and after partial demagnetization at high temperatures approached the level of sensitivity of the DSM magnetometer. Directions of magnetization found in sediments 100 m stratigraphically higher than the diabase sill are divergent. Specimen 75B (site LD31, Fig. 2*d*), for instance, shows a single magnetization of about 225°, +50° with respect to horizontal, and 265°, +05° with respect to bedding. The contact rocks and these remote sediments are lithologically different so a comparison of their intensities of magnetization is not meaningful. It is, however, evident that these results are indicative of a positive contact test (Everitt and Clegg 1962) demonstrating that the magnetization of the diabase is a reliable record of the Earth's field at, or soon after, the time of intrusion.

No systematic magnetic direction was identified with respect to the present horizontal, except perhaps for a recent Earth's field component which deflected NRM directions towards the lower hemisphere. Consequently no secondary magnetic overprint, comparable to that found in the Eskimo volcanics, has been isolated from the Haig intrusives. This was also true for the Flaherty volcanics. Fifty of the 91 samples collected have not yielded useful results. Many specimens behaved erratically during demagnetization and while the intrusive

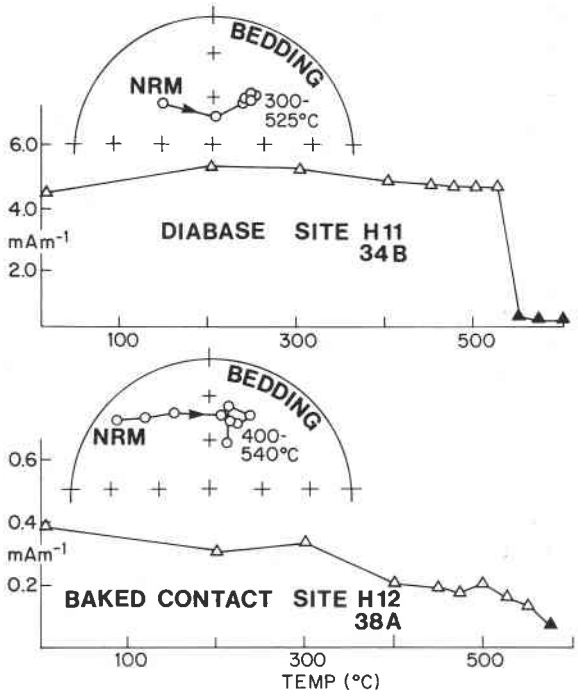



FIG. 6. Thermal demagnetization curves and stereographic projections of magnetization directions from a chilled margin and baked contact of the Haig intrusives.

rocks, in general, appeared more homogeneously magnetized than the volcanics, some specimens (i.e., from site H13, Fig. 5*a*) contained opposed components. This suggests that the specimens of intrusive rocks which *did not* yield useful results may contain mixed components. These latter specimens did, however, appear more altered than usual which may be relevant.

## Conclusions

### *Coexisting Opposed Magnetizations (COM): Origins and Implications for Polarity Magnetostratigraphy*

Throughout this study opposed (almost reversed) magnetizations have been found to coexist (e.g., specimens 32B, 39A, 40A, and 124A, Fig. 2; 15B, 152B, and 118B, Fig. 4; and 55a, Fig. 5). Although exact reversals are rare it should not be expected that such magnetizations can be perfectly isolated with present techniques, particularly if their stability spectra overlap. Consequently the term "opposed" has been adopted.

The coexistence of opposed magnetizations in igneous rocks is often interpreted as being indicative of some self-reversal mechanism having been operative during initial cooling (Nagata *et al.* 1952; Merrill and Grommé 1969). To investigate this, five specimens, all of which displayed COM during initial demagnetization, were heated to above their observed unblocking temperatures (600°C) and allowed to cool in the ambient magnetic field. Each specimen became magnetized in the Earth's field direction with intensities many orders of magnitude greater than their NRM intensities which supports the contention that their NRMs were composed of COM. Thermal demagnetization of these specimens revealed that only a single component of magnetization was present (Fig. 8). This experiment shows that if these specimens did undergo self-reversal of remanent magnetization when they originally cooled, then the causative mechanism has been destroyed presumably by the laboratory heating. While it is not possible to unequivocally reject nonreproducible self-reversal (Merrill and Grommé 1969) as the mechanism which produced the COM, it is of value to consider other mechanisms.

Low temperature oxidation has been considered a possible cause of self-reversal of magnetization in igneous rocks for many years (Verhoogen 1956, 1962; O'Reilly and Banerjee 1966). Lately, particular attention has been given to submarine basalts and many authors have reported apparent self-reversal features from samples of these basalts (Ozima and Larson 1968; Ade-Hall *et al.* 1971, 1973; Peterson and Bleil 1973). Although the cause(s) of self-reversed magnetizations in submarine basalts are little understood, there appears to be no reason to doubt their reality since their effects have been so widely observed. Conceivably, the COM described in this paper could be a manifestation of the same cause(s).

Whether nonreproducible self-reversal or a mechanism associated with low temperature oxidation (following closely in age the formation of the rock) is responsible for the COM, it should be noted that a reasonable approximation of the direction of the paleomagnetic field can still be determined, although its polarity would obviously be equivocal.

The magnetizations of the intrusives appeared to be less contaminated by COM than those of the volcanics. This can be accounted for by the deuteric alteration of the intrusives which occurred during cooling and imparted a resistance to further metamorphism (see Ade-Hall *et al.* 1971). The magnetizations of the volcanics display a greater scatter than do those of the intrusives (Fig. 7), reflecting their greater susceptibility to low temperature hydrothermal alteration and the COM that have resulted. Thus in analysing the paleomagnetic history of volcanics, it is suggested that it is important to consider not only obvious effects such as partial remagnetization associated with uplift (as in

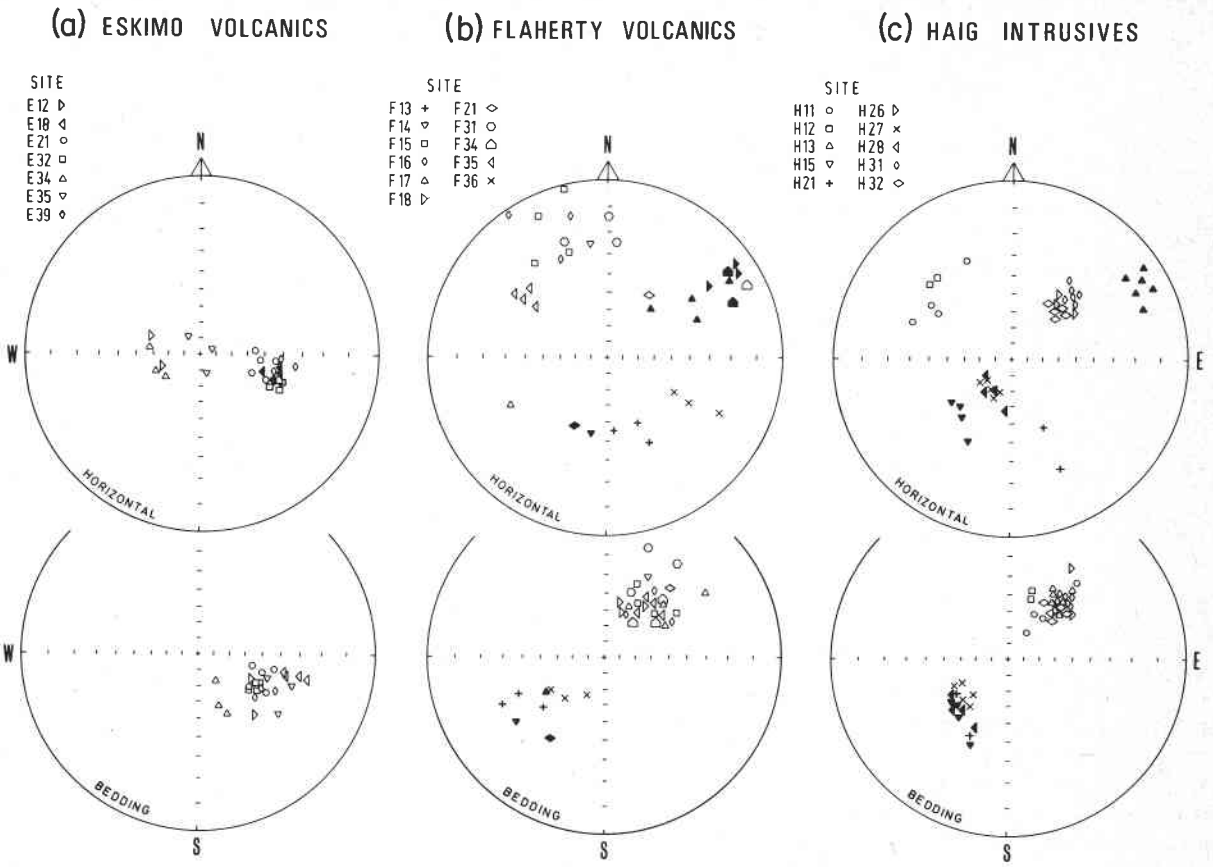



FIG. 7. Stereographic projections of cleaned magnetization directions from samples of the Eskimo and Flaherty volcanics and the Haig intrusives, comparing them with respect to present horizontal and with respect to bedding planes. The site mean directions are given in Tables 1, 3, and 4.

the Eskimo overprint), but also more subtle changes due to low grade metamorphism. Moreover, since present techniques are not adequate to unravel the complexities of every specimen's magnetizations, it ought not to be expected that sensible data can be gained from each sample, or even site, particularly if COM are carried by phases with overlapping stability spectra. Obviously, volcanics which have suffered alteration to the extent of those studied here cannot be considered for polarity correlation in magnetostratigraphic studies.

*Apparent Polar Wandering Path (APWP) for the Belcher Fold Belt*

The paleomagnetic pole positions determined from the stratigraphic sequence in the Belcher Islands reveal a systematic variation from the lower to the upper units (Fig. 9). There is sound evidence from stratigraphy, fold tests, and contact tests that this variation is due to the change in the paleomagnetic field during the times that these rock units were formed. Consequently, by invoking the dipole hypothesis, an apparent polar wandering path (APWP) can be constructed for the Belcher Islands. Moreover, it is reasonable to suppose that this APWP is representative for the whole Belcher Fold Belt and the Ungava Craton. The (anti-) pole positions from the Richmond Gulf andesites and basalts (Schwarz 1976) are in excellent agreement with the pole position of the Eskimo volcanics. These units are interpreted as stratigraphical equivalents by Dimroth *et al.* (1970), so their paleomagnetic correlation is indeed encouraging for future magnetostratigraphical studies. It is noted, however, that the magnetizations isolated by Schwarz were considered to be possibly secondary. The pole position given by Schwarz plotted close to the 1.2 Ga segment of the North American APWP, and in the absence of crucial field evidence, such as fold or contact tests, the magnetizations may well be treated as secondary. However, the

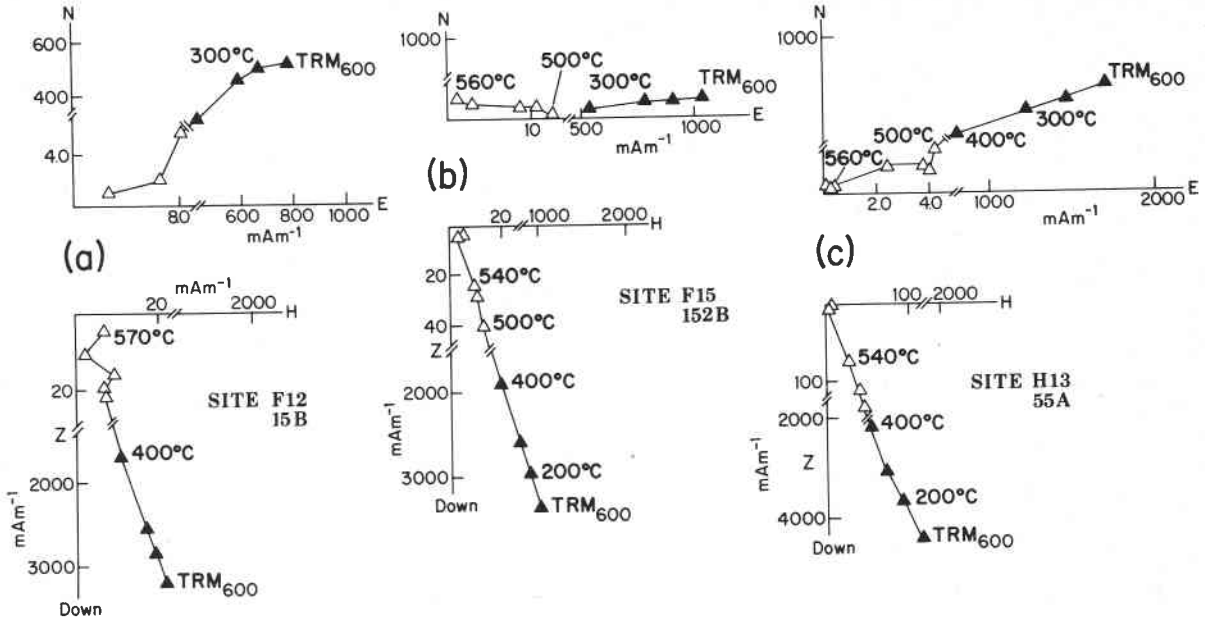



FIG. 8. Vector diagrams of thermal demagnetization behaviour of samples cooled to room temperature from 600°C in the ambient magnetic field. Note the change in scale of the axes.

independent data from the Belcher Islands, and the geological correlation of the rock units, lends strong support to their interpretation as primary directions.

The pole positions determined from the Flaherty volcanics and the associated Haig intrusives are indistinguishable from each other, but they are quite distinct from the pole position calculated from the primary magnetizations of the Eskimo volcanics. Thus, considerable motion with respect to the geomagnetic pole occurred between the times that the upper and lower volcanics were formed. The agreement of the poles from these units is consistent with their consanguinity, proposed on the basis of geological and geochemical reasoning (Dimroth *et al.* 1970).

The pole position calculated from the secondary magnetizations of the Eskimo volcanics is reasonably close to those of the Flaherty volcanics and the Haig intrusives (Fig. 9). This is interpreted as indicating that the sequence was folded soon after the outpourings of the Flaherty volcanics and final infilling of the basin, again supporting geological evidence (Dimroth *et al.* 1970) that suggests that the last sediments deposited were tectonically disturbed before consolidation.

Other paleomagnetic results from the geosyncline include those from the Labrador Trough (Park 1977), but because of the reconnaissance nature of that study Park found it difficult to give a definitive assessment of his results. With the new information now available from Richmond Gulf (Schwarz 1976, 1977) and the Belcher Islands it is opportune to review the Labrador Trough results in an effort to obtain a more coherent understanding of the overall development of the circum-Ungava geosyncline.

Park states "it is not possible at present to demonstrate from internal evidence that any of the magnetizations ... are primary." As such, most of the magnetizations were interpreted to be secondary, but their proposed age assignments were tentative.

In addition, some magnetizations were found in samples from only one or two sites while others have been subjectively selected to resolve different groups from an otherwise scattered population. These magnetizations might prove useful for future paleomagnetic work but, for the moment, it is preferred to restrict this discussion to those magnetizations that appear to be isolated with some degree of confidence.

The magnetizations which Park argued to be the oldest are those from the Castignon Complex (CT). In fact, these magnetizations were *tentatively* referred to the age of intrusion (approximately 1.9 Ga). However, these directions are very similar to the overprint directions found in the Cape Smith lavas (Fujiwara and Schwarz 1975), the Eskimo volcanics, and other formations (see Fig. 9), which suggests that the CT magnetizations could be 1.6–1.7 Ga old.

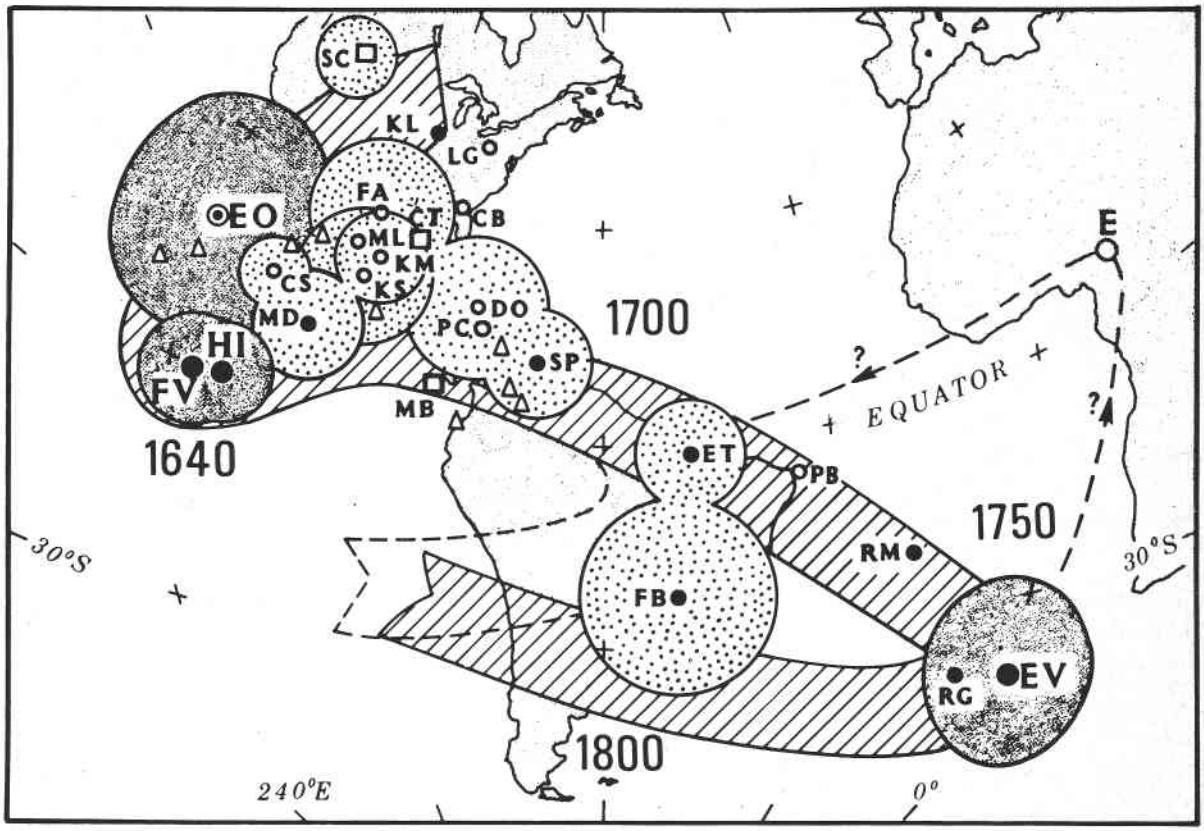



FIG. 9. Apparent polar wander path for the interval from about 1.8 to about 1.6 Ga ago. Solid symbols represent purportedly primary magnetizations while open symbols represent secondary magnetizations. The pole symbols are as follows: large symbols represent poles reported herein: CB, Charlton Bay Formation (Reid 1972); CS, Cape Smith basalts (Fujiwara and Schwarz 1975); CT, Castignon Complex (Park 1977); DO, Dogrib dikes (McGlynn and Irving 1975); E, "E" group (Roy and Lapointe 1976); EO, Eskimo overprint (this paper); ET, Et Then Group (Irving *et al.* 1972); EV, Eskimo volcanics (this paper); FA, Flin Flon greenstone belt (Park 1975); FB, Flin Flon greenstone belt (Park 1975); FV, Flaherty volcanics (this paper); HI, Haig intrusives (this paper); KL, Kaminak lamprophyre (Christie *et al.* 1975); KM, Kaminak dykes (Christie *et al.* 1975); KS, Kahochella Group (McMurray *et al.* 1973); LG, Gibraltar Formation (Reid 1972); MB, Menihek Formation overprint? (Park 1977); MD, Melville – Daly Bay metamorphics (Park 1973); ML, McLeod Formation (Reid 1972); PB, PC, Pearson Formation overprints (McGlynn and Irving 1978); RG, Richmond Gulf volcanics (Schwarz 1976); RM, Richmond Gulf – Manitounuk Island volcanics (Schwarz 1976); SC, Seward overprint (Park 1977); SP, Sparrow dikes (McGlynn *et al.* 1974). The open triangles represent poles from the Nonacho Group magnetizations of McGlynn *et al.* (1974) — excluding site 001 which forms part of the "E" group of Roy and Lapointe (1976). The dashed pole track joining pole E to the APWP is to indicate the uncertainty of this correlation. The dashed APWP from 1.8–1.7 Ga is the previously accepted APWP for North America (McGlynn and Irving 1978).

The presence of both normal and reversed magnetizations (MB) in the Menihek Formation lends some justification to selectively separating MA and MB directions, but at the same time would appear to favour the possibility that the MA magnetizations may be intermediate (or composite) and not distinct as suggested by Park (1977).

The MB magnetizations yield a pole position that is not significantly different from the CT pole position, again suggesting a similar age and probably a similar cause (i.e., the Hudsonian event(s)). The magnetizations removed during cleaning of samples from the Menihek Formation (MC) are not significantly different from the present geomagnetic field direction at the sampling sites and rather than attributing them to causes unknown it is suggested here that they have been acquired recently.

Samples from the Seward Formation yielded similar removed magnetizations (SC) to the MC magnetization, except for the presence of reversals which led Park (1977) to suggest that the SC pole belongs to track 3A. While this interpretation could be correct, the modified pole path given here suggests that the pole position could also be considered to belong to the young end of the path (Fig. 9). This would imply that these overprint magnetizations

are younger than most of the other overprints recognized in rocks affected by the Hudsonian event(s) and might be interpreted as indicating that the Labrador Trough underwent a protracted period of uplift from about 1.7–1.6 Ga.

While the above interpretations of the Labrador Trough magnetizations give a simpler picture than that presented by Park (1977) they are not unequivocal and await further data for verification. The ubiquity of the (Hudsonian) overprint magnetizations in the circum-Ungava geosyncline could represent a major difficulty in isolating primary magnetizations, particularly in rocks from the Labrador Trough.

*Apparent Polar Wandering Path (APWP) for North America 1.8–1.6 Ga*

Correlation of the APWP for the Belcher Fold Belt with the APWP for North America is not simple. The pole position from the Eskimo volcanics is remote from any other for this period except that of the Richmond Gulf andesites (Fig. 9). It requires a considerable APWP extension into the South Atlantic Ocean if North America was an integral unit. Consequently, it is possible to postulate a single-plate or a two-plate model, but with the paleomagnetic results currently available it is simply not possible to choose confidently which interpretation the data best fit. To test plate collision models (Gibb and Thomas 1976; Thomas and Gibb 1977) it would be necessary to determine APWPs from similarly aged rock sequences from the other structural provinces. Irving and McGlynn (1979) report results from the Coronation Geosyncline sequence in the Churchill Province and, although this sequence is older than that studied here, their conclusions generally support the single-plate model. It is worth noting that the paleomagnetic evidence to date favours one-plate models, not only for the cratonic nuclei of Laurentia (Hospers and Van Andel 1969; Roy and Lapointe 1976; McGlynn and Irving 1978) but also Australia (McElhinny and McWilliams 1977; Embleton 1978) and Africa (Piper *et al.* 1973; McElhinny and McWilliams 1977).

The paleomagnetic pole positions from the Flaherty volcanics and the Haig intrusives are in reasonable agreement with others from North America thought to be about this age (1.6 Ga). Thus, if any relative motion has occurred between the Ungava Craton and other parts of North America it would seem to have ceased by this time. It is noted too that the pole position from the Sparrow dikes (McGlynn *et al.* 1974) lies midway between the Flaherty–Haig poles and the Eskimo pole, in accordance with its age of about 1.7 Ga, which suggests that the Belcher Fold Belt APWP may be entirely conformable with that of North America, and the Eskimo and Richmond Gulf pole positions represent previously unknown North American pole positions.

The swath of pole positions from secondary magnetizations that have been isolated from North American rocks, and generally correlated with the Hudsonian event(s), appear to be about 1.65–1.7 Ga. The overprint pole from the Belcher Fold Belt is possibly slightly younger, at about 1.6 Ga, but is indistinguishable from the structurally uncorrected pole position from the Cape Smith Fold Belt (Fujiwara and Schwarz 1975). The Cape Smith basalts are probably overprinted (Schwarz 1976) and the coincidence of their pole with the Belcher overprint pole favours this interpretation, and indicates that the overprint was acquired at the same time as that of the Eskimo volcanics. Thus it is apparent that the circum-Ungava geosyncline was folded and uplifted at the same time, at about 1.6 Ga ago.

Paleomagnetic results from other rocks outside the geosyncline require mentioning. On the basis of a number of secondary magnetizations from the Nipissing diabase, and Huronian and Nonacho sediments, and purportedly primary magnetizations from the Stark Formation, Roy and Lapointe (1976) argue for an excursion towards the east into the North Atlantic Ocean for the time between ca.1.8–1.7 Ga. The secondary directions of magnetization have been observed by many authors (Patel 1971; Symons 1971; McGlynn *et al.* 1974; Bingham and Evans 1975; Roy and Lapointe 1976), so their reality would appear to be well established. Of less certainty, though, is their age. The secondary magnetizations could well represent a much younger pole path segment. The pole position denoted E (8°N, 14°E) in Fig. 9 is the mean of the group of E poles listed by Roy and Lapointe (1976, Table 8). The group is by no means tight, with $A_{95}$ = 26°, but led Roy and Lapointe (1976) to suggest an eastward extension of the North American APWP. The results herein generally support their notion.

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